

Diagnosing Direct-Substitution and One-Sided-Sign Errors

What direct substitution can and cannot settle:

- **A real number** — the function is continuous there and the substitution *is* the limit.
- **The form $0/0$** — indeterminate, not undefined. It reports that the substitution failed, nothing more; factor, rationalize, or combine fractions and try again.
- **Nonzero over zero** — the limit is infinite, and *which* infinity depends on the sign the denominator carries as x approaches from that side. Test the side, never assume it.

On every find-and-fix item, quote the line of the student's reasoning that fails, then redo the problem from that line on. Write ∞ or $-\infty$ where the limit is infinite, and write "does not exist" where the two sides disagree — those are different statements.

1. Evaluate $\lim_{x \rightarrow 3} (x^2 - 4x + 7)$, and state in one line why direct substitution is legitimate here.

Answer: _____

2. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$. Write down what direct substitution produces before you do anything else, then show the algebra that repairs it.

Answer: _____

3. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$. Substitution gives an indeterminate form; multiply by the conjugate to remove it. Give the exact value, not a decimal approximation.

Answer: _____

4. Find and fix the mistake. A student writes

$$\lim_{x \rightarrow 2} \frac{1}{x-2} = \infty$$

arguing that the denominator shrinks to zero, so the fraction grows without bound. Evaluate the two one-sided limits separately, then state what the two-sided limit is and why.

Answer: _____

5. Find and fix the mistake. A student writes

$$\lim_{x \rightarrow 0} \frac{|x|}{x} = 1$$

saying that $|x|$ and x “cancel”. Evaluate the limit from the left and from the right, and state what the two-sided limit is.

Answer: _____

6. Find and fix the mistake. A student claims that $\lim_{x \rightarrow 4^+} \frac{x-7}{x-4} = \infty$, because “approaching from the right always gives positive infinity”. Determine both one-sided limits at $x = 4$ and explain what the student’s rule left out.

Answer: _____

7. Find and fix the mistake. Let

$$f(x) = \begin{cases} x^2 + k, & x < 3 \\ 5x - 1, & x \geq 3 \end{cases}$$

A student is asked which value of k makes f continuous at $x = 3$ and answers $k = 14$, “because $5(3) - 1 = 14$ ”. Write the continuity condition as an equation, solve it for k , and give the common value of the two one-sided limits.

Answer: _____

8. Explain. Write a short paragraph for a classmate who believes that substitution either gives the answer or shows the limit does not exist. Say what the form $0/0$ actually reports and what you do next; say what nonzero-over-zero reports; and explain why two infinite one-sided limits with opposite signs mean the two-sided limit does not exist.